

		MIS 1/MIS 2	MIS 5e/MIS 6	MIS 7a- c/MIS 7d	MIS 7e/MIS 8	MIS 9e/MIS 10	MIS 11c/MIS 12	MIS 13a/MIS 14	MIS 15a/MIS 15b	MIS 15e/MIS 16	MIS 17/MIS 18	MIS 19c/MIS 20	MIS 21e/MIS 22	MIS 25/MIS 26	MIS 27/MIS 28	MIS 29/MIS 30	MIS 31/MIS 32	MIS 33/MIS 34	MIS 35/MIS 36	MIS 37/MIS 38	MIS 39/MIS 40	MIS 41/MIS 42	MIS 43/MIS 44	MIS 45/MIS 46	MIS 47/MIS 48
Benthic $\delta^{18}\text{O}$ (‰)	Lisiecki & Raymo (2005)	1.84	1.88	0.96	1.19	1.65	1.97	1.08	0.94	1.59	1.25	1.26	1.27	1.28	0.56	0.97	1.08	0.83	0.90	1.03	0.79	0.60	0.85	0.91	1.07
	Ahn et al. (2017)	1.72	1.85	0.94	1.26	1.67	1.95	1.08	0.96	1.68	1.04	1.28	1.32	1.22	0.68	0.79	1.04	0.83	1.11	1.10	0.71	0.74	0.98	0.77	1.10
	Zhou et al. (2026) - BIGSTACK_mixed	1.65	1.73	0.89	1.12	1.58	1.85	0.89	0.78	1.51	0.91	1.12	1.24	1.06	0.49	0.74	0.81	0.62	0.97	1.11	0.62	0.61	0.86	0.42	0.98
	Zhou et al. (2026) - BIGSTACK_mixedA	1.72	1.90	1.01	1.28	1.73	1.98	1.06	0.87	1.70	1.06	1.34	1.42	1.11	0.59	0.89	0.93	1.01	1.12	1.37	0.70	0.78	1.01	0.80	1.30
	Zhou et al. (2026) - BIGSTACK_mixedP	1.60	1.54	0.82	1.05	1.39	1.77	0.82	0.66	1.27	0.92	1.04	1.22	1.01	0.50	0.60	0.96	0.60	0.88	1.02	0.58	0.53	0.81	0.70	1.00
ΔSST (°C)	Clark et al. (2024) - Global Mean ΔSST stack	3.43	3.71	1.37	2.42	3.18	4.47	1.85	0.49	2.29	1.94	3.16	2.96	1.16	0.50	1.23	1.87	1.57	1.75	2.30	0.68	1.18	2.09	0.66	1.78
	Clark et al. (2024) - North Atlantic ΔSST stack	6.18	6.99	2.69	3.95	7.52	8.16	3.13	0.95	4.11	6.70	6.72	4.25	2.78	3.97	1.92	6.03	3.74	4.17	5.20	1.89	3.76	4.78	1.09	4.15
	Clark et al. (2024) - Tropical Atlantic ΔSST stack	3.51	3.05	1.81	2.68	3.67	4.86	2.62	1.31	3.79	2.58	3.11	4.65	1.27	0.75	2.85	1.32	2.28	3.02	2.44	0.20	1.60	1.63	2.15	2.94
	Clark et al. (2024) - South Atlantic ΔSST stack	4.48	5.61	1.79	4.05	5.37	6.95	3.51	0.74	3.59	3.14	3.40	3.67	5.31	2.47	3.98	4.92	4.44	5.90	5.37	4.43	1.58	1.87	3.17	4.41
	Clark et al. (2024) - North Pacific ΔSST stack	3.59	7.46	3.63	3.77	5.46	6.06	3.58	1.48	3.75	2.82	5.02	5.26	1.41	1.52	0.77	3.42	0.88	4.00	2.49	2.70	1.24	4.57	1.40	3.92
	Clark et al. (2024) - Tropical Pacific ΔSST stack	2.67	2.44	0.87	1.78	2.26	2.18	0.19	0.53	1.09	1.93	1.02	1.89	0.78	0.86	0.97	1.36	1.37	0.97	1.18	0.45	0.79	1.66	0.60	0.92
	Clark et al. (2024) - South Pacific ΔSST stack	4.56	6.20	2.34	3.49	4.51	6.27	3.53	0.31	6.48	4.32	5.94	4.93	3.02	2.43	5.60	3.30	2.76	2.25	1.86	3.31	4.34	4.05	3.10	3.23
	Clark et al. (2024) - Tropical ΔSST stack	2.69	2.54	1.12	2.01	2.45	2.47	0.58	0.52	1.28	1.76	1.70	2.24	0.70	0.57	1.11	1.14	1.48	1.36	1.32	0.35	0.57	1.63	0.54	1.26
	Herbert et al. (2010)	3.08	2.67	1.39	2.65	3.06	3.40	1.64	1.12	1.83	2.30	2.48	3.44	1.78	0.79	1.84	1.81	1.71	1.87	2.40	0.52	1.61	2.59	0.77	1.42
ΔMOT (°C)	Clark et al. (2025)	3.43	3.70	1.36	2.41	3.19	4.47	1.85	0.48	2.28	1.93	3.17	2.96	1.08	0.45	1.06	1.56	1.26	1.30	1.62	0.45	0.75	1.27	0.39	0.96
Mean loess size (normalised)	Ding et al. (2002)	0.47	0.94	0.53	0.39	0.74	0.65	0.20	0.21	0.79	0.49	0.36	1.09	0.28	0.12	0.16	0.13	0.48	0.40	1.28	0.32	0.27	0.23	0.17	0.36
Mean size (μm)	Sun et al. (2021)	27.07	30.21	23.93	19.56	25.97	24.02	18.42	17.50	41.53	23.13	19.89	36.71	12.38	14.52	10.05	18.40	34.25	13.32	21.44	15.67	11.28	15.30	15.55	17.00
	Sun et al. (2019)	20.97	21.54	26.93	14.09	26.60	21.64	14.01	11.25	24.47	16.37	16.88	31.68	18.33	12.57	12.54	15.31	20.77	21.63	31.84	16.13	8.13	14.05	17.81	11.75
Inorganic loess carbonates $\delta^{13}\text{C}$ (‰)	Sun et al. (2019)	5.63	7.25	5.10	5.14	5.10	5.46	4.87	4.21	5.57	6.14	5.00	5.39	3.76	3.36	4.91	4.25	3.73	4.98	5.05	4.11	3.16	4.29	5.17	4.27
MAP (mm/day)	Sun et al. (2019)	0.33	0.48	0.32	0.41	0.38	0.37	0.30	0.31	0.35	0.34	0.32	0.36	0.39	0.31	0.27	0.42	0.17	0.33	0.27	0.21	0.21	0.39	0.19	0.36
MAT (°C)	Sun et al. (2019)	2.60	3.20	2.20	2.50	2.40	2.70	1.60	2.00	2.40	2.00	0.80	1.60	2.30	2.10	1.30	1.60	1.40	1.40	1.70	1.30	1.10	1.90	1.10	2.40
	Zhao et al. (2021)	3.43	2.85	1.25	0.93	1.04	1.39	2.36	1.00	2.04	2.43	1.79	2.38	2.08	1.29	1.42	3.12	1.46	1.88	2.02	1.13	2.89	1.70	2.22	3.41
AP (excl. Graminea and aquatic/semi aquatic taxa) (%)	Tzedakis et al. (2006)	97.39	95.41	91.74	96.26	92.29	100.00	82.34	89.01	93.85	86.89	66.97	88.31	89.65	31.22	54.76	91.11	68.61	59.25	83.15	69.79	67.48	nan	nan	nan
AP (excl. pinus) (%)	Donders et al. (2021)	72.73	83.30	80.02	77.13	80.17	79.31	84.77	70.58	77.96	53.29	73.83	83.38	75.67	47.12	59.48	63.31	74.11	61.51	52.27	60.01	78.38	nan	nan	nan
AP (%)	Zhao et al. (2020)	48.87	84.42	41.08	48.34	62.21	77.85	73.89	47.71	73.73	84.64	63.18	86.95	64.40	87.29	71.16	73.53	71.26	73.83	73.19	80.78	55.64	55.18	72.98	82.55
AP (%), excl. Quercus & Alnus	Torres et al. (2013)	nan	57.75	17.55	41.82	43.68	37.22	33.48	19.09	31.70	51.79	41.03	39.37	30.41	37.23	24.74	23.25	25.49	27.75	39.78	20.72	24.06	16.12	17.75	34.80
Si/Ti	Melles et al. (2012)	0.42	0.42	0.42	0.48	0.98	2.00	0.46	0.66	0.69	0.95	0.71	0.56	0.82	0.39	0.48	1.49	0.45	0.45	0.67	0.64	0.60	0.75	0.58	0.42
Biogenic Silicia (BioSi %)	Prokopenko et al. (2006)	nan	41.30	37.60	29.60	39.80	43.70	32.10	44.20	41.00	38.80	34.20	44.00	40.90	13.70	24.00	43.30	6.10	36.00	48.20	17.20	28.00	16.00	36.10	34.00
Log Ca/K	Sun et al. (2021)	5.82	6.64	5.45	5.35	5.67	6.70	5.92	3.73	7.12	6.74	6.69	6.94	6.15	4.61	6.13	6.91	5.96	6.76	7.02	5.42	5.32	nan	nan	nan